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Impact of hydraulic characteristics on irrigation canals failure in Jember, Indonesia

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Abstract. Irrigation system revetment damage, caused by external flow, disrupts efficient water distribution and reduces agricultural output. This study investigated the hydraulic characteristics of these canals as a preventative measure against such damage. The analysis was conducted one-dimensionally (1D) using two flow approaches: steady and unsteady flow. The research took place in Jember Regency, East Java Province, Indonesia, examining 20 straight canal sections damaged with about 20 meters long at Antirogo, Rembangan, Bintoro, and Jumerto district. Model validation was conducted using current meter measurements. Under steady flow conditions, flow rates ranged from 0.012 to 0.215 m³/s average 0.106 m³/s; with average upstream and downstream velocities of 0.250 m/s and 0.300 m/s. Water surface elevations were between 0.150 and 0.440 m, and the average slope was 0.002 m/m. The Froude number 0.100–0.490 consistently represent subcritical flow. Statistical analysis confirmed that hydraulic characteristics significantly impact canal damage, as evidenced by relationships between upstream or downstream flow velocities, water level and Froude number. Hydraulic modeling with HEC-RAS is an effective tool for evaluating individual canal hydraulics, supporting adaptive maintenance strategies based on actual canal conditions.

1. Introduction

Irrigation systems are critical infrastructure that play a vital role in supporting food security and enhancing agricultural productivity. Their primary function is to ensure the availability and efficient distribution of water to agricultural land. However, in many regions, these systems face serious challenges due to damage that leads to a decline in water distribution performance. Such damage includes canal wall erosion, sedimentation, cracking, and structural failures, which result in uneven and inefficient water delivery (1), (2). In the context of irrigation infrastructure management, Indonesian Government Regulation (PP) No. 20/2006 about Irrigation emphasizes the importance of integrated asset management, encompassing inventory, planning, implementation, and evaluation processes (3). Unfortunately, repair efforts are often not based on a technical understanding of the hydraulic characteristics of water flow that directly interact with the structure.

Although numerous studies have examined the physical condition of irrigation systems, a gap in the literature reveals that there is still a lack of research specifically linking the degree of physical damage to hydraulic parameters such as flow velocity, water surface profile, and the



Froude number. These parameters are critical in determining flow behaviour and the potential for further structural damage (4).

In this context, hydraulic modeling approaches have been widely employed to evaluate the behaviour of open channel flows. Software such as HEC-RAS 1D is considered effective for modeling hydraulic characteristics, including determining water surface profiles and the Froude number, which can indicate flow types: subcritical, supercritical, or transitional (4), (5), (6), (7). The value of the Froude number is crucial because it enables the identification of flow regimes (laminar, transitional, and supercritical), and its influence plays a significant role in the design process and in anticipating potential structural damage (5).

Based on this premise, the innovation of this research lies in developing a hydraulic modeling approach using HEC-RAS 1D to examine the relationship between the degree of irrigation system damage and key hydraulic parameters. This approach allows for the identification of critical points based on simulation results, which can serve as the basis for more targeted rehabilitation planning. Furthermore, this study also takes into account local geological conditions, such as the degree of rock weathering, which also affects structural stability (5).

2. Materials and Method

2.1. Study Area

This study was conducted in Jember Regency, East Java Province, with 20 survey points selected in conditions ranging from severe to moderate damage (Figure 1). Each surveyed location covered a ± 20 -meter section with a straight geometry. The surveyed sites were located in the Antirogo, Rembangan, Bintoro, and Jumerto irrigation areas.

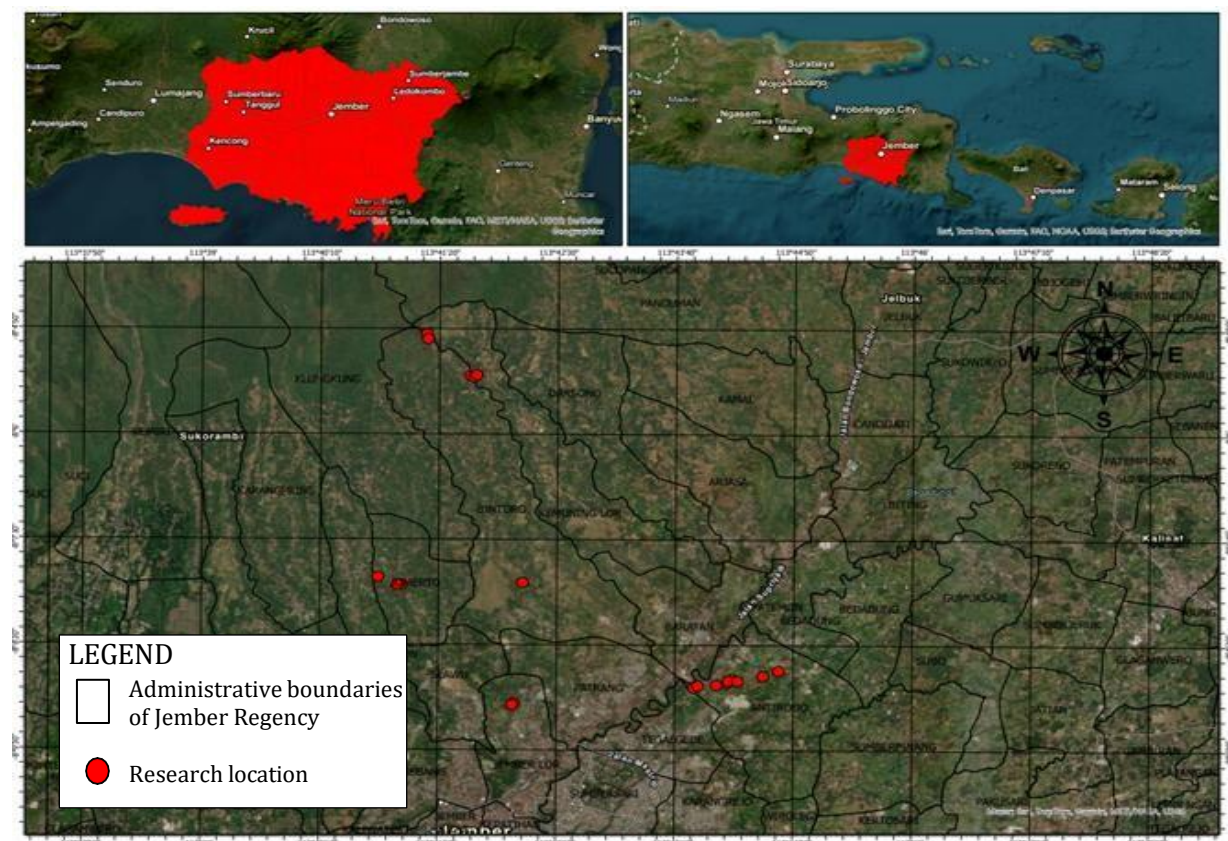


Figure 1. Study area map

2.2 Data Collection

The data used in this study consists of historical records, irrigation discharge, flow velocity, geometric profiles, and stone masonry construction design criteria (Table 1).

Table 1. Data Source

Data	Data Resources	Data Function
Damage Data	Public Works and Water Resources Agency (DPU BM-SDA) of Jember Regency	To calibrate and validate the HEC-RAS model
Velocity	Field survey using a Flowatch current meter	To determine the discharge rate for HEC-RAS model calibration
Water Level Height	Field measurements of water surface elevation (m)	As input data for HEC-RAS modeling
Channel Geometry	Field survey of slope (m/m) and dimensions (m)	
Channel Material Type	Estimated Manning's roughness coefficient	As input data for HEC-RAS modeling

2.3 Research Stages

The methodology used in this research and in the modeling of irrigation channels in Jember consists of two main stages. The first stage involves the collection and processing of channel data. The second stage includes the development of a hydraulic model using HEC-RAS. The flowchart about research stages can see in Figure 2.

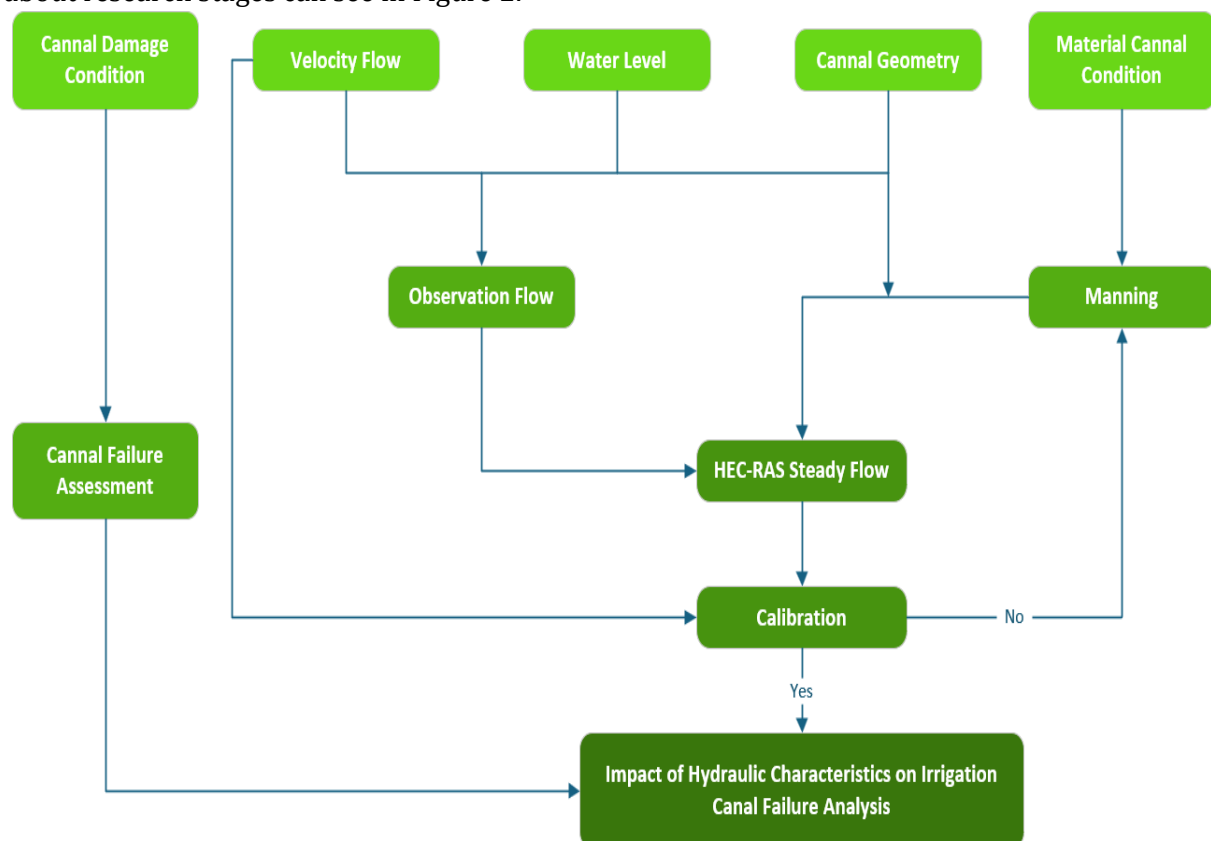


Figure 2. Research flowchart

2.4 Determination Infrastructure Damage

The assessment of damage to irrigation canals is based on the Ministry of Public Works and Housing Regulation No. 23/PRT/M/2015 of 2015, which contains the Implementation Guidelines for Assessing the Performance Index of Irrigation Systems (Table 2), including a table describing each condition observed in the field (8).

Table 2. Performance index of irrigation systems

No	Indicator	Score	Excellent Condition (90%–100%)	Minor Condition (80%–<90%)	Moderate Condition (60%–<80%)	Severe Condition (<60%)
1	Channel profile capacity vs design plan	50	Meets design capacity	Minor reduction ≤20%	Some channels reduced >40%	All channels reduced >40%
2	Illegal tapping, leakage, and efficiency	40	No illegal tapping/leakage, efficiency >90%	Minor tapping/leakage, efficiency 80–90%	Several tapping/leakages, efficiency 60–70%	Many tapping/leakages, efficiency <60%
3	Sedimentation or erosion	10	No effect >10%	Minor effect 10–20%	Significant effect 20–40%	Major effect >40%

According to the Ministry of Public Works and Housing Regulation No. 12/PRT/M/2015, the damage scale can be determined using the Condition Weight Value (NKB) formula as $NKB = \sum(\text{Score} \times \text{Condition Value})/100$. The classification criteria are: $NKB > 90\%$: Good condition, $NKB > 80-90\%$: Minor damage, $NKB > 60-80\%$: Moderate damage and $NKB < 60\%$: Severe damage

2.5 Flow

Field measurements were conducted using a current meter to obtain flow velocity. The measurement location was divided into two points, namely the upstream and downstream sections. A rope or cable was stretched across the designated cross-section to serve as the location for discharge measurement. Each section consisted of vertical points spaced at intervals of 1/15 of the channel width. At each vertical point, the water depth was measured, and the flow velocity was determined using the six-tenths method (measurement at a depth of 0.600 from the water surface). The current meter was lowered to the designated measurement point according to the predetermined depth, and the flow velocity value at that point was displayed on the current meter (9). The discharge at each point was then calculated using the following Equation 1.

$$Q = \frac{1}{n} AR^{\frac{2}{3}} S^{\frac{1}{2}} \quad \text{Eq. 1}$$

where Q is Flow discharge (m^3/s), n is Manning's roughness coefficient, R is Hydraulic radius of the channel, A is Cross-sectional area of the channel and S is Channel slope (10). The relationship between flow velocity and structural damage is a fundamental concept in hydraulics and river engineering, wherein high flow velocities can trigger erosion (11).

2.6 Manning

The Manning's roughness coefficient (n) accounts for the resistance to flow caused by the characteristics of the channel surface. Its value depends on the material and surface condition (12). The Manning's n for stone masonry ranges from 0.015 to 0.030, whereas for earthen channels it ranges from 0.026 to 0.050 (13). The Manning coefficient also influences several hydraulic parameters, such as flow velocity (v), discharge (Q), and the vertical velocity profile (14).

2.7 HEC-RAS Steady Flow

The modeling system component is designed to calculate water surface profiles for gradually varied steady flow. This system is capable of handling a complete network of channels, dendritic systems, or a single river reach. The steady flow component can model water surface profiles for subcritical, supercritical, and mixed flow regimes (15). In this study, steady flow analysis is employed to obtain flow velocity values that correspond to field conditions, as well as to determine the condition of the channel under full-flow conditions. It is also used to derive other variables such as the Froude number. The equations used in the steady flow analysis are as follows:

One-Dimensional Energy, Equation 2:

$$z_1 + y_1 + \alpha_1 \frac{v_1^2}{2g} = z_2 + y_2 + \alpha_2 \frac{v_2^2}{2g} + h_L \quad \text{Eq. 2}$$

where z is Channel bed elevation (m), y is Water depth (m), α is Velocity Head Coefficient, v is Flow velocity (m/s), g is Gravity (9.810 m/s²), and h_L is Total Energy Loss (m).

Total Energy Loss (h_L) caused by friction (friction loss) in Equation 3 and 4:

$$h_f = S_f \times L \quad \text{Eq. 3}$$

$$S_f = \frac{n^2 v^2}{R^3} \quad \text{Eq. 4}$$

where S_f is energy gradeline slope, n is Manning's coefficient, R is hydraulic radius (m) = A/P , and L is distance between cross-sections (m).

Froude Number in Equation 5:

$$Fr = \frac{v}{\sqrt{gDh}} \quad \text{Eq. 5}$$

where Dh is Hydraulic depth (Area / water surface width).

2.8 Calibration

HEC-HMS calculates and presents four summary statistics to evaluate model performance against observations. These statistics include the Nash–Sutcliffe Efficiency (NSE), the Ratio of the Root Mean Square Error to the Standard Deviation of the Observations (RSR), and the Percent Bias (PBIAS) (16). The corresponding equations are as follows:

$$NSE = 1 - \left[\frac{\sum_{i=1}^n (Y_i^{obs} - Y_i^{sim})^2}{\sum_{i=1}^n (Y_i^{obs} - \bar{Y}_{obs})^2} \right] \quad RSR = \frac{RMSE}{\sigma_0} = \frac{\sqrt{\sum_{i=1}^n (Y_i^{obs} - Y_i^{sim})^2}}{\sqrt{\sum_{i=1}^n (Y_i^{obs} - \bar{Y}_{obs})^2}} \quad PBIAS = \left[\frac{\sum_{i=1}^n (Y_i^{sim} - Y_i^{obs}) \times (100)}{\sum_{i=1}^n (Y_i^{obs})} \right] \quad \text{Eq. 6}$$

where Y_i^{sim} is i -th simulated value, Y_i^{obs} is i -th observation, $\bar{Y}_{obs}$ is average of observed data, n is mean of the observed data, and σ is total number of observations (15).

The validation focused on two primary parameters, namely flow velocity at the upstream and downstream points of 20 canal segments. Field data were obtained through measurements using

a current meter, while the simulation results were derived from the steady flow output of the HEC-RAS model.

2.9 Ordinal Logistic Regression

Ordinal Logistic Regression is a statistical analysis method used when the dependent variable is in the form of ordered categories (ordinal), such as “minor”, “moderate”, and “severe” (17). This model is also commonly referred to as the *proportional odds model* because it assumes that the effect of the independent variables on the likelihood of moving to a higher category is constant across all category levels (18).

To understand how to interpret the coefficients, we first introduce some notation and review the concepts involved in ordinal logistic regression. Let Y be an ordinal outcome with J categories. Then $P(Y \leq j)$ denotes the cumulative probability that Y is less than or equal to a given category j , where $j = 1, \dots, J - 1$. The odds of being less than or equal to a given category can be defined as follows Equation 7:

$$\frac{P(Y \leq j)}{P(Y > j)} \quad \text{Eq. 7}$$

To $j = 1, \dots, J - 1$ while $P(Y > J) = 0$, the division by zero is undefined. Log odds are also referred to as the logit:

$$\log \frac{P(Y \leq j)}{P(Y > j)} = \text{logit}(P(Y \leq j)) \quad \text{Eq. 8}$$

In R, the Proportional Odds Logistic Regression (POLR) model is parameterized as (19):

$$\text{logit}(P(Y \leq j)) = \beta_{j0} - \eta_1 x_1 - \dots - \eta_p x_p \quad \text{Eq. 9}$$

In this context, the relationship between ordinal logistic regression and hydraulic characteristics emerges when the data are classified into specific categories, for example, canal embankment damage categorized as “minor,” “moderate,” and “severe.” In this study, the independent variables consist of physical parameters such as discharge, flow velocity, Manning’s coefficient, slope friction, Froude number, and Reynolds number.

3. Results and Discussion

3.1 Canal Failure Assessment

Based on Table 3 about observation condition assessment of 20 irrigation canals revealed three distinct categories of damage severity.

Table 3. Observation condition assessment

No	Canal Location	Documentation	Assessment Result	Category
1	Amsar		Capacity reduced >40% in some sections	Moderate
2	Budar		Many leaks and >40% sediment/erosion	Severe

No	Canal Location	Documentation	Assessment Result	Category
3	Komina		Minor leaks; 10–20% sediment/erosion	Minor
4	Antirogo		Capacity reduced >40% in some sections	Moderate
5	Haji karim		Minor capacity change $\leq 20\%$	Minor
6	Karman		Capacity reduced >40%; 20–40% sediment/erosion	Moderate
7	Sukorejo		Minor leaks; 20–40% sediment/erosion	Moderate
8	Bedadung		Capacity reduced >40% in some sections	Moderate
9	Makam		Capacity reduced >40% in some sections	Moderate
10	Muktisari		Capacity reduced >40% in all sections	Severe
11	Kenari		Minor leaks	Minor

No	Canal Location	Documentation	Assessment Result	Category
12	Curahkates		Many leaks; efficiency <60%	Severe
13	Batu		Minor capacity change ≤20%	Minor
14	Langon		Capacity reduced >40%; minor illegal tapping/leaks	Moderate
15	Bintoro		Capacity reduced >40%; 20–40% sediment/erosion	Moderate
16	Rayap		Capacity reduced >40% in all sections; >40% sediment/erosion	Severe
17	Perbal		Minor capacity change ≤20%; 10–20% sediment/erosion	Minor
18	Sembah		Capacity reduced >40%; >40% sediment/erosion	Severe
19	Petung		Minor leaks; 20–40% sediment/erosion	Moderate
20	Arah		Minor capacity change ≤20%; 10–20% sediment/erosion	Moderate

The analysis began with the Moderate category, which dominated the sample, accounting for 10 canals (50%). These canals typically exhibited capacity reductions exceeding 40% or sedimentation/erosion in the range of 20–40%. Such conditions align with previous studies demonstrating that even moderate sedimentation levels can substantially reduce hydraulic efficiency and water delivery performance (20). The Severe category was identified in 5 canals (25%). These canals suffered from significant leakage, extensive sedimentation or erosion exceeding 40%, and overall reductions in cross-sectional capacity.

This category represents critical conditions requiring urgent rehabilitation to prevent functional collapse. Prior research has shown that sedimentation beyond 40% can obstruct water conveyance and lead to partial structural failure if left unaddressed. Finally, the Minor category also included 5 canals (25%), marked by limited leakage or light sedimentation (10–20%). These canals remained operationally efficient, requiring only routine maintenance. This finding supports studies noting that lined or well-maintained canals experience minimal leakage and retain high water distribution efficiency.

3.2 Calibration of the HEC-RAS Model Using Field Data

Based on the HEC-RAS modeling, the calibration results can be grouped into three categories as shown in Table 4. Group 1 includes canals with very high accuracy (NSE = 1; RSR = 0; PBIAS = 0), such as Haji Karim, Sukorejo, Bedadung, Makam, Muktisari, Kenari, and Bintoro. Group 2 represents overestimated simulations with positive PBIAS values, including Amsar (0.9931; 1.05%; 0.083), Budar (0.700; 2.08%; 0.707), Antirogo (0.944; 2.38%; 0.236), Batu (0.988; 1.18%; 0.102), Arah (0.959; 0%; 0.200), and Karman (0.750; 7.14%; 0.354). Group 3 consists of underestimated simulations with negative PBIAS values, such as Curahkates (0.944; -1.72%; 0.236), Langan (0.973; -4.17%; 0.164), Rayap (0.957; -2.04%; 0.210), Perbal (0.973; -0.92%; 0.160), Sembah (0.960; -5.26%; 0.200), and Petung (0.846; -9.52%; 0.390).

Overall, the calibration results demonstrate that the HEC-RAS model performed with high accuracy and consistently represented field conditions. Therefore, the model can be considered reliable for further hydraulic simulations and hydraulic characteristic on canal.

3.3 Evaluation of HEC-RAS Simulation Results (Steady Flow)

HEC-RAS simulations obtain a detailed of their hydraulic conditions has shown at Bar Chart in Figure 3. The analysis Manning's, which range from 0.032–0.050. The highest value was recorded at Curahkates, indicating a rougher surface, while the lowest was at Langan, showing relatively smooth conditions. Flow discharge varied between 0.011 m³/s in Sembah and 0.215 m³/s in Amsar, where higher flow was generally associated with greater velocity and depth. Velocity showed that some canals accelerated from upstream to downstream, such as Langan (0.350 m/s to 0.540 m/s), while Sembah maintain stable velocities (0.100 m/s). Water Level ranged from 0.150–0.440 m, with Budar nearly full, while Bedadung and Sembah had lower levels with small discharges. Friction slope values were highest in Batu and Langan, reflecting significant energy loss due to rough surfaces or rapid flows. All Froude numbers were below 1 confirming subcritical flow, though Langan and Batu approached the critical threshold. Meanwhile, Reynolds numbers all exceeded 2,000, confirming turbulent flow, with the highest at Amsar and the lowest at Sembah.

3.4 Hydraulic characteristic model selection

Model selection was conducted using the Akaike Information Criterion (AIC) to balance the quality of model fit (goodness of fit) with model complexity (parsimony). At the initial stage, a set of candidate models was constructed that encompassed all possible combinations of available

hydraulic characteristics, resulting in a total of 233 regression models that could be mathematically evaluated. For each model, the hydraulic characteristics were estimated using the maximum likelihood method, and the log-likelihood value was obtained.

All models were then ranked according to their AIC values, and ΔAIC was computed as the difference between the AIC of a given model and the lowest AIC observed. ΔAIC was used to assess model competitiveness: $\Delta AIC = 0$ indicates the best model; $\Delta AIC < 2$ indicates alternative models that are still competitive; ΔAIC between 4–7 indicates weakened model support; and $\Delta AIC > 10$ indicates models that are considerably worse than the best model.

The model selection results showed that the best model was obtained with the combination of the hydraulic characteristics Flow (flow_m3_s) Downstream velocity (v_d_stream_m_s), Water level (water_level_m), and Froude number (angka_froude), with an AIC value of 48.416 ($\Delta AIC \approx 0.290$). Two other models with $\Delta AIC < 2$, namely the model including Flow, Upstream velocity, Downstream velocity, Water Level (AIC = 46.970; $\Delta AIC \approx 0.670$) and the single-variable model with Downstream Velocity (AIC = 46.465; $\Delta AIC \approx 0.160$), were still considered competitive but were not selected as the primary model because their balance between complexity and predictive ability was less optimal compared to the best model.

Alternative models with slightly higher AIC values were also considered; however, according to (21), models with $\Delta AIC < 2$ are still competitive, while those with $\Delta AIC \geq 4$ receive considerably weaker support. Therefore, the final selection determined that the combination of *Flow + Downstream Velocity + Water Level + Froude Number* represented the best model, as it provided the most optimal balance between goodness of fit and reasonable model complexity.

Table 4. Calibration Model

Canal Location	Calibration Statistics			Canal Location	Calibration Statistics		
	NSE	PBIAS	RSR		NSE	PBIAS	RSR
Amsar	0.993	1.05%	0.083	Kenari	1	0	0
Budar	0.700	2.08%	0.707	Curahkates	0.944	-1.72%	0.236
Komina	0.920	1.64%	0.283	Batu	0.988	1.18%	0.109
Antirogo	0.944	2.38%	0.236	Langon	0.973	-4.17%	0.164
Haji Karim	1	0	0	Bintoro	1	0	0
Karman	0.750	7.14%	0.354	Rayap	0.957	-2.04%	0.210
Sukorejo	1	0	0	Perbal	0.973	-0.92%	0.160
Bedadung	1	0	0	Sembah	0.960	-5.26%	0.200
Makam	1	0	0	Petung	0.846	-9.52%	0.390
Muktisari	1	0	0	Arah	0.959	0%	0.200

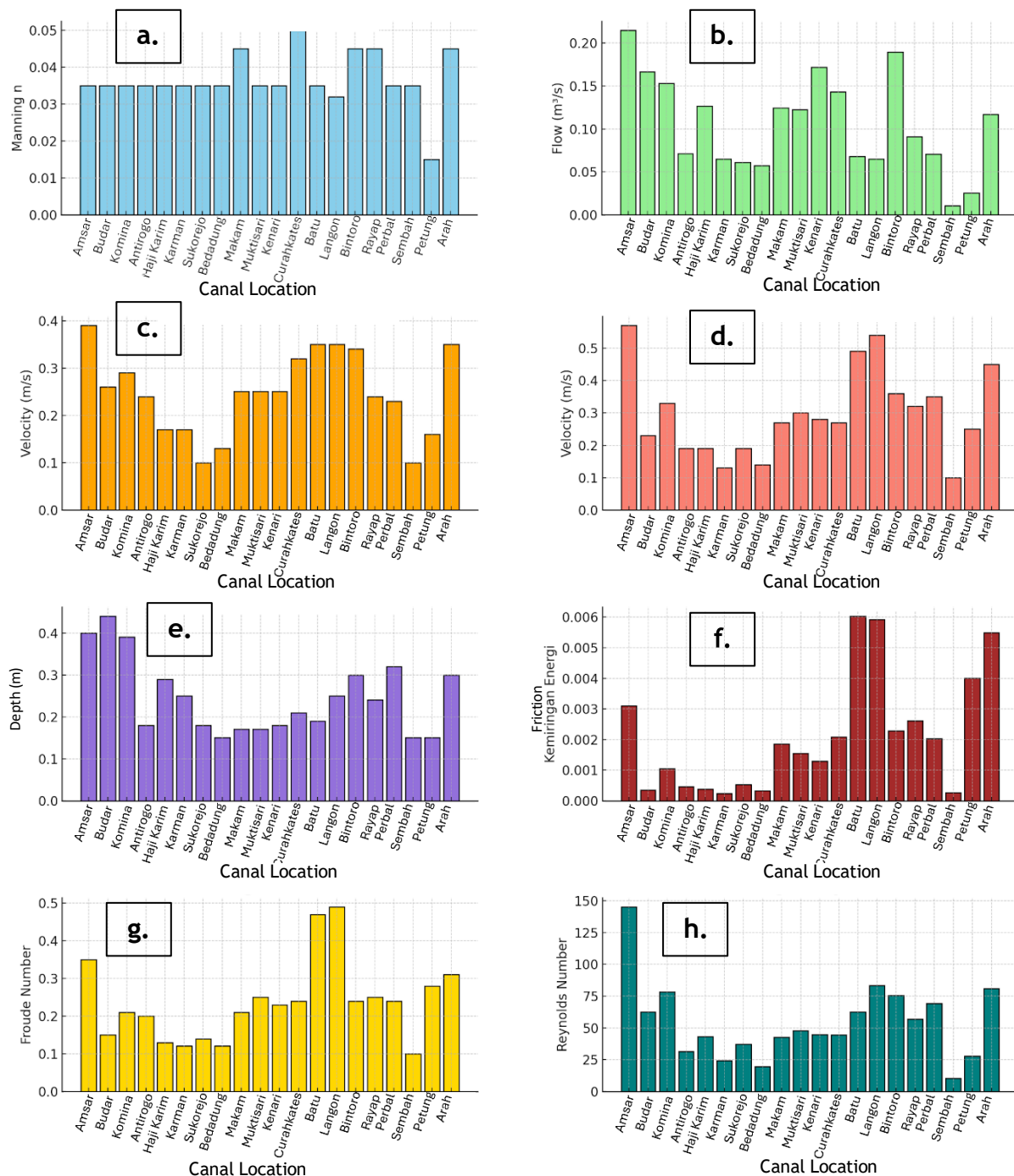


Figure 3. The figure presents eight bar charts showing hydraulic characteristics of irrigation channels at various locations, including: a). Manning's, b). flow discharge, c). upstream velocities, d). downstream velocities, e). water level, f). friction slope, and the dimensionless parameters g). Froude number and h). Reynolds number.

3.5 Impact of Hydraulic Characteristic on Irrigation Canal

The analysis employed POLR to examine how hydraulic factors influence the ordinal condition levels of irrigation channels—classified as Minor, Moderate, and Severe. The process began with data collection and organization, followed by model construction to evaluate four predictors:

Flow, Froude number, Velocity Downstream, and Water Level. The estimation results revealed distinct patterns. Flow had a negative coefficient (-13.670), indicating that higher discharge tends to reduce the likelihood of severe damage, although this effect was not statistically significant ($t = -1.149$). In contrast, the Froude number showed a large positive coefficient (25.220), suggesting that greater flow dynamics increase the odds of higher damage severity, with a near-significant effect ($t = 1.626$). Similarly, Velocity Downstream (12.34, $t = 1.914$) and Water Level (15.040, $t = 1.732$) both had positive coefficients, implying that increases in these parameters tend to raise the probability of channels shifting toward more severe damage. Threshold parameters were estimated at -0.413 for the transition from Minor to Moderate, and 2.286 for the transition from Moderate to severe, highlighting those severe conditions require more extreme hydraulic stresses. Although statistical significance was limited, the results indicate that downstream velocity, Froude number, and water level increase risk severity, while higher flow may mitigate it. These insights provide a preliminary basis for prioritizing maintenance and rehabilitation efforts in irrigation channel management.

`polr(formula = Condition ~ Flow + Froude + `V D Stream` + `Water Level`, data = Irrigation)`

Coefficients:

	Value	Std. Error	t value
Flow	-13.670	11.896	-1.149
Froude	25.220	15.516	1.626
`V D Stream`	12.340	6.447	1.914
`Water Level`	15.040	8.687	1.732

Intercepts:

	Value	Std. Error	t value
Minor Moderate	-0.413	1.587	-0.260
Moderate Severe	2.286	1.673	1.366
Residual Deviance:	36.416	AIC:	48.416

Form Table 5, the analysis identified downstream velocity as the factor most closely associated with irrigation channel damage severity, with the smallest p-value (0.056). This suggests that increases in downstream velocity significantly raise the likelihood of transitioning from minor to more severe conditions. Technically, higher velocity elevates the flow's kinetic energy, increasing basal shear stress on channel beds and banks. When shear exceeds the resistance capacity of surface materials such as gravel or fine soils, erosion occurs, particularly in unprotected embankments catastrophic failure or riprap (22). The Froude number ($p = 0.104$) also showed a positive influence on damage severity. High values indicate more energetic flows, approaching supercritical conditions, which heighten structural stress. This aligns with findings on structural scour, where elevated Froude numbers enhance turbulence, generate vortices around structures, and increase bed shear stress (23). Similarly, water level ($p = 0.083$) exerted a near-significant positive effect. Higher water surfaces increase hydrostatic pressure, while rapid drawdown after prolonged rainfall can raise pore water pressure, weaken slope stability, and trigger cascading failures (24). In contrast, flow showed a negative but non-significant effect ($p = 0.251$), suggesting that larger discharges within adequately sized channels do not necessarily intensify damage. Based on these findings, the analysis concludes that downstream velocity, Froude number, and water level are the variables most closely associated with an increased risk of damage.

Table 5. Proportional Odds Logistic Regression Result

Variable	Estimate	t-value	p-value	Remarks
Flow	-13.670	-1.149	0.251	Not significant
Froude	25.220	1.626	0.104	Nearly significant
Velocity Downstream	12.340	1.914	0.056	Nearly significant (close to p=0.05)
Water Level	15.040	1.732	0.083	Nearly significant

4. Conclusion

This study began with field data collection from several irrigation canals in Jember Regency, including measurements of discharge, upstream and downstream velocity, water level, bed slope, and physical canal conditions. These data were then used to calibrate the HEC-RAS hydraulic model, producing hydraulic parameters that accurately represented real field conditions. The evaluation of 20 irrigation canals through HEC-RAS steady flow simulations, correlation analysis, condition assessment, and statistical modeling provides an integrated understanding of canal performance and vulnerability. The hydraulic simulations confirmed subcritical but turbulent flow across all canals, with site-specific variations in discharge, velocity, and roughness significantly shaping hydraulic behavior. Correlation analysis further emphasized that discharge and velocity are the primary drivers of canal dynamics, strongly influencing Froude and Reynolds numbers, while water level and friction slope showed weaker relationships due to local geometric and structural factors.

The condition assessment revealed that Moderate damage was the most common category (50%), followed by Severe (25%) and Minor (25%). These findings highlight sedimentation, leakage, and erosion as key issues that substantially reduce capacity and efficiency, particularly in canals requiring urgent rehabilitation to prevent structural failure.

The Proportional Odds Logistic Regression (POLR) results identified downstream velocity, Froude number, and water level as the hydraulic variables most associated with increasing damage severity, although none reached full statistical significance. Downstream velocity emerged as the closest to significance, underscoring its critical role in erosion and embankment instability. In contrast, higher discharges tended to reduce risk, likely due to flow accommodation in adequately sized channels.

Overall, the study concludes that effective irrigation canal management should prioritize controlling downstream velocity, mitigating high Froude number conditions, and managing water level fluctuations, while focusing rehabilitation on severely affected canals and maintaining moderately damaged ones to sustain long-term hydraulic efficiency.

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